

Make feasibility a first-class search signal

A controlled physical-design experiment about where EDA time goes.

FlowGuard asks a practical question: when every RTL-to-GDS run costs real minutes, can a search policy spend fewer of them on configurations that were never physically viable?

$$EI \times P(\text{feasible})$$

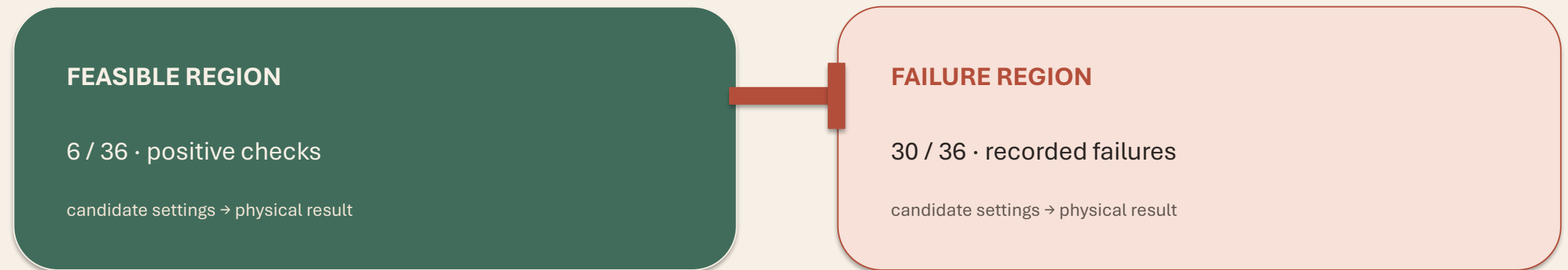
same budget / different allocation

The claim is about failed-run compute, not a promise of better final QoR.

One crossbar. Two physical outcomes.

The 19.9 ns crossbar-v2 boundary is a 36-configuration oracle, not the older stress benchmark.

Same crossbar, fixed constraints, different outcomes: six feasible configurations and thirty failures.



19.9 ns

Quality plus feasibility, separated on purpose.

FlowGuard learns what is good only where a run is physically valid—and learns validity from every attempted trial.

QUALITY MODEL

**GP → expected
improvement**

trained on feasible trials only

FEASIBILITY MODEL

RF → P(feasible)

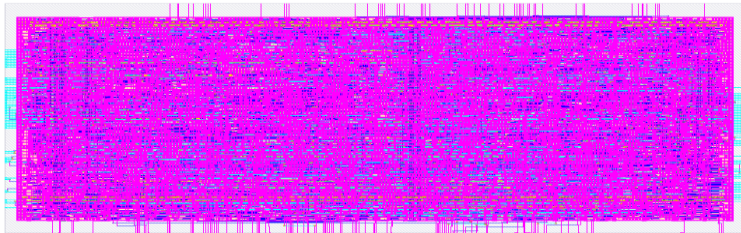
trained on all attempted trials



$$\text{acquisition} = \text{EI} \times \text{P(feasible)}$$

One decision. One authentic physical artifact.

Replay selects cb36_002; the layout image comes from its separate banked physical run.



cb36_002 · SHA-256 35fbbd7e...a12dc5de

separate-run artifact

FlowGuard-Calibrated · seed 11 · call 10

selected cb36_002

EI 0.0 × P(feasible) 0.08 · rank 1 / 27

feasible · QoR -0.093068 · setup +0.190903 ns

DRC 0 · routing 100 · LVS/signoff pass

Freeze the comparison before interpreting it.

A matched 24-evaluation replay across 13 seeds and three methods.

Design	crossbar_datapath_v2
Clock	19.9 ns (fixed)
Pool	36 configurations · 6 / 30 oracle
Budget	24 evaluations: 8 shared init + 16 adaptive
Methods	Vanilla · FlowGuard · EI-only

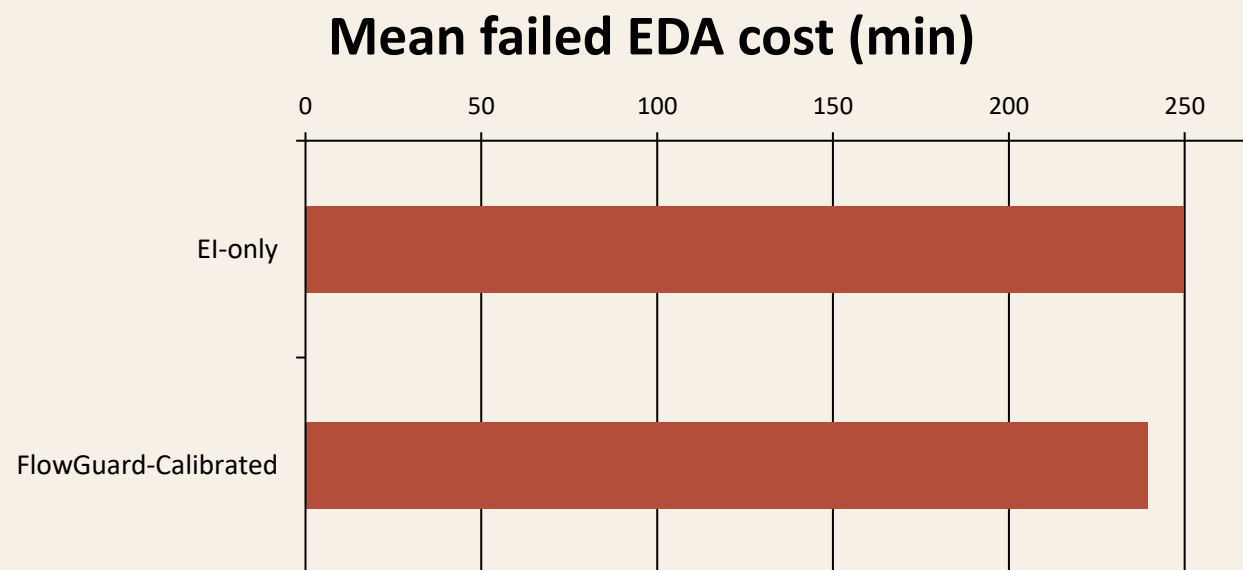
No replay EDA: the bundle replays committed outcomes; physical metrics are banked oracle results.

HEADLINE RESULT

Feasibility weighting reduced failed-run compute by 4.1%

FlowGuard versus matched EI-only search across 13 replay seeds

10.28 fewer minutes spent on failed configurations per replay, on average



-4.11%

derived: $(14994.8 - 14378.0) / 14994.8$

same final QoR and best-call outcome; total cost slightly higher

Derived from verified failed means 14994.8 s and 14378.0 s: 616.8 s = 10.28 min. Editable zero-based chart; sequential EDA-runtime estimate.

Three methods. Vanilla spends least.

Across 13 seeds, all three tie final QoR; cost separates failed and total runtime means.

Method	Mean feasible	Failed EDA mean (s)	Total EDA mean (s)	Final QoR	Best-call
Vanilla	6.00	13046.8	17230.7	-0.093068	baseline
FlowGuard	4.38	14378.0	17307.3	-0.093068	= El-only
El-only	3.38	14994.8	17272.8	-0.093068	= FlowGuard

Vanilla is lowest on both failed EDA mean and total EDA mean; final QoR ties three ways.

Means from final_evidence_bundle/03_recomputed_aggregates.json; feasible counts are seed means.

Reproduce the decision, not just the layout.

Crossbar replay is an oracle/cache process, not a new physical campaign.

Older TPE / layout image

physical artifact $\neq$ decision record

keep separate from crossbar evidence

FlowGuard / decision record

seed · call · hash · scores · rank

cb36_002 replay record is traceable

Limitations: no standalone combined-13 verification log; replay/oracle outcomes attach banked physical runs; older stress evidence is excluded.

Reproduce the replay. Then run the hardware.

The crossbar bundle makes the decision traceable; the next step is independent physical confirmation.

Run ``python3 experiments/replay_demo.py``, inspect `cb36_002`, then verify the banked artifact independently.

REPRODUCE
then falsify

1 freeze

2 run

3 inspect

Next test: independently reproduce the 13-seed tie and validate the separate physical artifact chain.